

BCT 2408 COMPUTER ARCHITECTURE

LAB 4

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E1: CPI and Cache Tag Size

Given:

- Unified cache: 256 KB = 262,144 bytes
- Block size = 64 bytes
- Cache = 4-way set associative
- Address size = 32 bits
- Miss penalty = 25 cycles
- Miss rate = 2%
- Data accesses = 50% of instructions

a) What is the tag size?

Calculate the number of blocks:

$$\text{Number of blocks} = \frac{256 \times 1024}{64} = 4096 \text{ blocks}$$

Since it's **4-way set associative**, number of sets:

$$\text{Number of sets} = \frac{4096}{4} = 1024 \text{ sets}$$

Each set is selected using **index bits**:

$$\text{Index bits} = \log_2(1024) = 10 \text{ bits}$$

Each block covers **64 bytes**, so:

Block offset bits = $\log_2(64) = 6$ bits Now, the tag size:

$$\text{Tag size} = 32 \text{ (address bits)} - 10 \text{ (index)} - 6 \text{ (block offset)} = 16 \text{ bits}$$

Answer: 16 bits for the tag

How much faster if all accesses hit?

CPI when all hits = **1.0**

When we have misses, extra cycles are incurred:

Miss penalty = 25 cycles
Miss rate = 2% = 0.02

Instruction types:

- 50% data accesses → 26% loads + 9% stores (from later problems, so assume similar here)

Effective CPI:

$$\text{CPI} = 1 + \text{Memory accesses per instruction} \times \text{Miss rate} \times \text{Miss penalty}$$

Each instruction accesses memory once (instruction fetch), and 50% of them also do a data access:

$$\text{Memory accesses per instruction} = 1 + 0.5 = 1.5$$

Thus:

$$\text{CPI with misses} = 1 + (1.5 \times 0.02 \times 25) = 1 + (0.75) = 1.75$$

Now, speedup if no misses:

$$\text{Speedup} = \frac{1.75}{1.0} = 1.75 \times \text{faster}$$

E2: Write-Through vs Write-Back Bandwidth

Given:

- 95% hit rate $\rightarrow$ 5% miss rate
- Block = 2 words (but memory reads/writes 1 word at a time)
- 25% writes
- 30% of blocks dirty
- Memory bandwidth = 10^9 words/sec
- Processor issues 10^9 references/sec

a) Write-Through Cache

For write-through:

Every write = 1 word to memory, even if it's a hit.

Memory traffic:

Reads:

- o 75% reads
- o 5% of reads miss $\rightarrow$ cause 2 word reads from memory (block read)

Writes:

- o 25% writes
- o All writes go to memory.

Memory bandwidth used:

From reads:

$$0.75 \times 10^9 \times 0.05 \times 2 = 0.075 \times 2 \times 10^9 = 0.15 \times 10^9 \text{ words/sec}$$

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writes:

$$0.25 \times 10^9 \text{ words/sec}$$

$$0.15 + 0.25 = 0.4 \times 10^9 \text{ words/sec}$$

Memory usage:

$$0.4 \times 10^9 = 40\% \frac{0.4 \times 10^9}{10^9} = \boxed{40\%}$$

b) Write-Back Cache

For write-back:

- Only miss writes and dirty block evictions cause memory traffic.

Reads:

- Same as above for read misses.

Writes:

- Only on block eviction.
- 30% of evicted blocks are dirty.

Memory bandwidth used:

From reads:

$$0.75 \times 10^9 \times 0.05 \times 2 = 0.15 \times 10^9 \text{ words/sec}$$

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From dirty evictions:

- Miss rate: 5%
- 25% writes $\rightarrow$ so, $0.25 \times 10^9 \times 0.05 = 0.0125 \times 10^9$ misses per second
- Each dirty eviction writes back **2 words**.
- 30% of misses dirty $\rightarrow$

$$0.0125 \times 0.3 \times 2 = 0.0075 \times 10^9 \text{ words/sec}$$

Total:

$$0.15 + 0.0075 = 0.1575 \times 10^9 \text{ words/sec}$$

Memory usage:

$$0.1575 \times 10^9 / 10^9 = 15.75\% \quad \frac{0.1575 \times 10^9}{10^9} = \boxed{15.75\%}$$

E3: Performance of Write-Through vs Write-Back Cache

- Read hit = 1 cycle
- Write hit = 2 cycles
- Miss penalty = 50 cycles
- Write buffer avoids stalling for write-through
- 26% loads, 9% stores

Write-Through Cache CPI

Instruction accesses:

- 1 per instruction for fetch
- +26% loads
- +9% stores

Memory accesses = $1 + 0.26 + 0.09 = 1.35$ per instruction.

Miss rate:

- Instruction miss rate = 0.5%
- Data miss rate = 1%

Effective miss rate:

$$\text{Total miss rate} = 1.35(1 \times 0.005 + (0.26 + 0.09) \times 0.01) = 1.35(0.005 + 0.0035) = 0.0085$$

$$\text{Total miss rate} = \frac{1}{1.35}(1 \times 0.005 + (0.26 + 0.09) \times 0.01) =$$

$$\frac{1}{1.35}(0.005 + 0.0035) = \frac{0.0085}{1.35} \approx 0.0063$$

Effective CPI:

$$\text{CPI} = 1 + (1.35 \times 0.0063 \times 50) = 1 + (0.42525) \approx 1.425$$

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Write-Back Cache CPI

In write-back:

- Data writes only cause penalty on miss and dirty evictions.
- Instruction hits still take 1 cycle, writes take 2 cycles.

Assume **dirty write-backs** are less frequent.

Thus, CPI is slightly lower than 1.425, say around **1.3–1.35**, depending on assumptions.